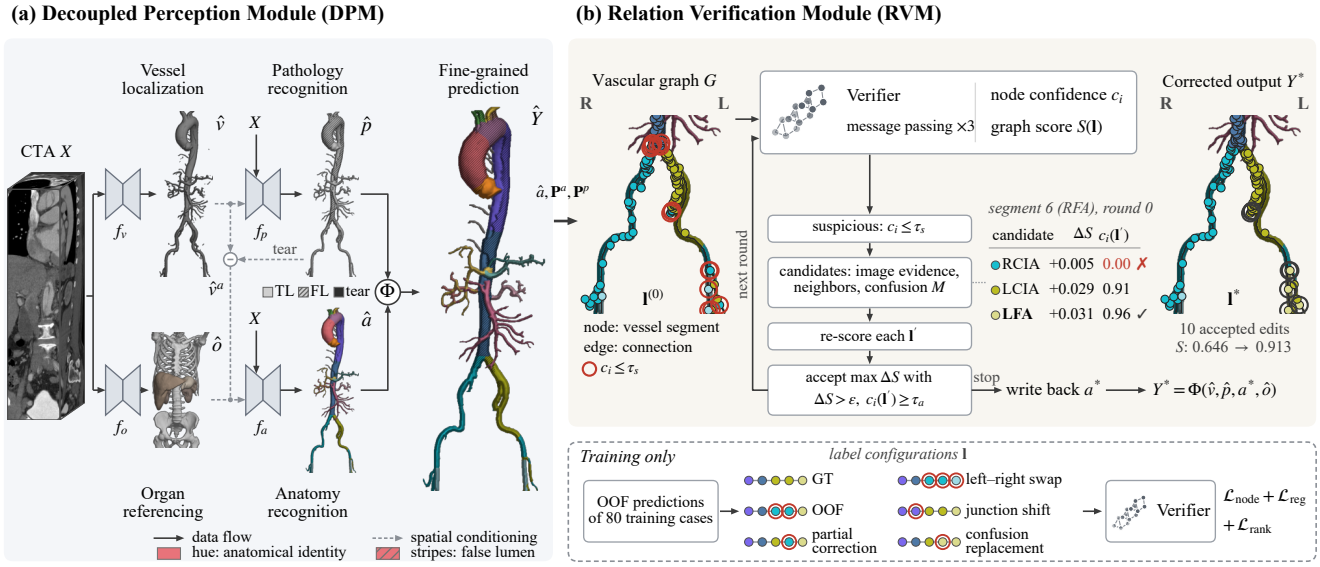

(a) Decoupled Perception Module (DPM)
(b) Relation Verification Module (RVM)
Vessel localization
Pathology recognition
Fine-grained prediction
CTA X
tear
TL FL tear
Organ referencing
Anatomy recognition
data flow
hue: anatomical identity
spatial conditioning
stripes: false lumen
Vascular graph G
R
L
node: vessel segment
edge: connection
Verifier
message passing ×3
node confidence c_i
graph score $S(\mathbf{l})$
next round
suspicious: $c_i \le \tau_s$
candidates: image evidence, neighbors, confusion M
re-score each $\mathbf{l}'$
accept max ΔS with
$\Delta S > \varepsilon$, $c_i(\mathbf{l}') \ge \tau_a$
stop
write back a^*
$Y^* = \Phi(\hat{v}, \hat{p}, a^*, \hat{o})$
segment 6 (RFA), round 0
candidate ΔS $c_i(\mathbf{l}')$
RCIA +0.005 0.00 ✗
LCIA +0.029 0.91
LFA +0.031 0.96 ✓
Corrected output Y^*
10 accepted edits
S: 0.646 → 0.913
Training only
label configurations $\mathbf{l}$
OOF predictions of 80 training cases
GT
OOF
partial correction
left–right swap
junction shift
confusion replacement
Verifier
$\mathcal{L}_{\text{node}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{rank}}$